

Capital Gains Realization Model

A Minimal Specification

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1 Purpose

The model tracks how a capital-gains reform changes the stock of unrealized gains over time, and how that changed stock feeds back into realizations and revenue.

The central object is not the total stock of unrealized gains. The central object is the reform-minus-baseline difference:

$$\Delta G_{a,t,k} = G_{a,t,k}^S - G_{a,t,k}^B.$$

Here a is age, t is year, and k is asset class. The asset classes are equities, pass-throughs, primary homes, other homes, and real estate funds.

This setup is useful because the simulator is a repeated cross-section. It does not literally follow the same taxpayer over time. The model instead uses age-by-year cells as synthetic cohorts. A person aged a in year t who survives becomes part of age $a + 1$ in year $t + 1$. If the person dies, the model either forgives, transfers, or realizes the embedded gain depending on the policy regime.

The intuition is a bathtub. Unrealized gains sit in the tub. Realizations drain the tub. Death either drains the tub, routes the water to heirs, or turns it into immediate tax revenue. A tax reform changes both the drain rate and the death treatment, so the stock begins to diverge from baseline.

2 Baseline Objects

Each cell (a, t, k) starts with baseline quantities from the simulator:

- $G_{a,t,k}^B$: baseline unrealized taxable gains.
- $r_{a,t,k}^B$: baseline realization rate.
- $R_{a,t,k}^B$: observed baseline realized gains.
- $m_{a,t}$: mortality rate for the tax unit cell.
- $\tau_{a,t}$: capital-gains marginal tax rate for the cell.

The baseline realization rate is

$$r_{a,t,k}^B = \frac{R_{a,t,k}^B}{G_{a,t,k}^B},$$

with the usual sparse-cell guard when $G_{a,t,k}^B = 0$.

Two allocation matrices move gains across age cells:

$$\sum_h \omega_{a \rightarrow h,t} = 1 \quad \text{for every decedent age } a.$$

$A_{a \rightarrow h,t}$ moves survivors from age a to destination age h . $\omega_{a \rightarrow h,t}$ routes inherited carryover gains from decedents at age a to heirs at age h .

For one-year ages with a topcode at $a_{\max}$,

$$A_{a \rightarrow h,t} = \begin{cases} 1 & \text{if } a < a_{\max} \text{ and } h = a + 1, \\ 1 & \text{if } a = a_{\max} \text{ and } h = a_{\max}, \\ 0 & \text{otherwise.} \end{cases}$$

3 Realization Behavior

Realizations come from two distinct channels. Some sales are weakly tax-elastic turnover: business exits, life-event-driven home sales, divorce-induced asset divisions, forced liquidations, and portfolio rebalancing not driven by tax timing. The holder's flexibility on these sales is small, so the hazard governing them is approximately policy-invariant. The remaining sales are discretionary timing decisions that respond to the tax wedge.

The total realization rate is the sum of both channels:

$$r_{a,t,k} = \lambda_{a,t,k}^I + r_{a,t,k}^V.$$

$\lambda_{a,t,k}^I$ is the turnover hazard: the portion of baseline realization treated as weakly tax-elastic and policy-invariant. $r_{a,t,k}^V$ is the voluntary, tax-elastic rate.

λ^I is not an alternative realization rate. It is the policy-invariant turnover hazard used to price the value of waiting. The model's total realization rate is still r^S ; the split only says that part of baseline realization is weakly tax-elastic and part is discretionary. Both pieces are real sales that show up in revenue.

3.1 Voluntary Realization Decision

Normalize to one dollar of embedded gain. A representative holder in cell (a, t, k) chooses a voluntary realization fraction r^V :

$$\max_{r^V \in [0, 1 - \lambda_{a,t,k}^I]} b_{a,k}(r^V) - P_{a,t,k} r^V.$$

$b_{a,k}(r^V)$ is the non-tax reason to *voluntarily* realize: discretionary rebalancing, profit-taking, tax-loss harvesting on adjacent positions. $P_{a,t,k}$ is the effective tax price of selling now instead of waiting.

Use the marginal-benefit form

$$b'_{a,k}(r^V) = \mu_{a,k} - \frac{1}{\eta_k} \log r^V.$$

The first-order condition is

$$\log r_{a,t,k}^{V*} = \eta_k(\mu_{a,k} - P_{a,t,k}).$$

The intercept $\mu_{a,k}$ does not need to be separately estimated. It is absorbed by anchoring the voluntary channel to its observed baseline value:

$$r_{a,t,k}^{V,B} = r_{a,t,k}^B - \lambda_{a,t,k}^I.$$

The unconstrained exponential response can exceed the feasible upper bound $1 - \lambda_{a,t,k}^I$ under large rate cuts. Cap the scenario voluntary rate accordingly:

$$r_{a,t,k}^{V,S} = \min \left\{ 1 - \lambda_{a,t,k}^I, r_{a,t,k}^{V,B} \exp[-\eta_k (P_{a,t,k}^S - P_{a,t,k}^B)] \right\}.$$

The cap binds only when the unconstrained exponential would overshoot the cell's available unrealized stock. The total scenario realization rate is

$$r_{a,t,k}^S = \lambda_{a,t,k}^I + \min \left\{ 1 - \lambda_{a,t,k}^I, (r_{a,t,k}^B - \lambda_{a,t,k}^I) \exp[-\eta_k (P_{a,t,k}^S - P_{a,t,k}^B)] \right\}.$$

If the reform is the baseline, $P^S = P^B$, the exponential is one, the cap does not bind, and $r^S = r^B$ by construction.

The two limits clarify the structure. When $\lambda_{a,t,k}^I = 0$, the model collapses to a single-channel specification with full elasticity on r^B – the previous spec. When $\lambda_{a,t,k}^I = r_{a,t,k}^B$, the entire baseline rate is turnover and $r^S = r^B$ regardless of the tax wedge; the cell is fully inelastic. Asset classes with high turnover share are mechanically less responsive to rate changes, matching the empirical pattern that housing has a lower realization elasticity than liquid equities.

3.2 Effective Tax Price

Selling today costs τ_t per dollar of gain. Holding avoids that tax today, but the dollar may still leave the holder's control later through turnover, carryover at death, or deemed realization at death. The effective tax price is the tax due now minus the discounted expected tax due later through these largely policy-invariant paths:

$$P_{a,t,k}(c) = \tau_t - \sum_{j=1}^{\infty} \beta^j s_{a,k,j} \tau_{t+j} - \sum_{j=1}^{\infty} \beta^j d_{a,j} c \tau_{t+j}.$$

The terms are:

- β : annual discount factor, default near 0.96.
- $s_{a,k,j}$: probability of *turnover* realization in future year $t+j$, conditional on still holding at t .

- $d_{a,j}$: probability of death in future year $t+j$ before the asset has otherwise exited the holder's control.
- c : share of the embedded tax burden that survives death and is internalized by the current holder.

Voluntary future realizations do not appear in the price. Future voluntary choices are themselves endogenous responses to the future tax wedge, not exogenous events with a fixed probability. Including them here would create a fixed-point dependence between P and r^V that would have to be solved iteratively. Restricting the price to turnover paths and death keeps the bracket genuinely exogenous and avoids the recursion.

The death-regime values of c are:

$$c = \begin{cases} 0 & \text{step-up basis,} \\ \theta \in (0, 1] & \text{carryover basis,} \\ 1 & \text{deemed realization at death.} \end{cases}$$

Under step-up, death wipes out the gain, so death has no future-tax cost to the holder. Under deemed realization, death fully taxes the gain. Under carryover, θ says how much the holder cares about the heir's eventual tax bill.

If expected future rates are constant, $\tau_{t+j} = \tau_t$, the price factors into a rate and a bracket:

$$P_{a,t,k}(c) = \tau_t (1 - M_{a,k}(c)),$$

where

$$M_{a,k}(c) = \sum_{j=1}^{\infty} \beta^j s_{a,k,j} + c \sum_{j=1}^{\infty} \beta^j d_{a,j}.$$

The bracket $1 - M_{a,k}(c)$ is the part of today's tax rate that really bites. If future taxes are likely anyway, the extra cost of selling today is smaller. That is why removing step-up can raise the level of realizations but lower the sensitivity to tax-rate changes.

3.3 Future Turnover and Death Probabilities

The probabilities s and d come from a competing-risks process. Each year, a holder either turns over the asset, dies, or keeps holding.

The probability that the asset is still held at the start of year $t+j$ is

$$S_{a,k,j} = \prod_{i=0}^{j-1} (1 - \lambda_{a+i,t+i,k}^I - m_{a+i,t+i}), \quad S_{a,k,0} = 1.$$

Then

$$s_{a,k,j} = S_{a,k,j} \lambda_{a+j,t+j,k}^I, \quad d_{a,j} = S_{a,k,j} m_{a+j,t+j}.$$

The model usually truncates the infinite sums after a long horizon because discounting, mortality, and turnover make the tail negligible.

3.4 Elasticity Implied by the Bracket and Voluntary Share

Under constant-rate expectations,

$$\frac{\partial P_{a,t,k}}{\partial \tau_t} = 1 - M_{a,k}(c).$$

The voluntary-channel semi-elasticity is therefore

$$\frac{\partial \log r_{a,t,k}^{V,S}}{\partial \tau_t} = -\eta_k (1 - M_{a,k}(c)).$$

Evaluated at baseline ($P^S = P^B$), the total-rate semi-elasticity is the voluntary semi-elasticity scaled by the voluntary share of total baseline realizations:

$$\left. \frac{\partial \log r_{a,t,k}^S}{\partial \tau_t} \right|_{P^S=P^B} = -\eta_k (1 - M_{a,k}(c)) \frac{r_{a,t,k}^{V,B}}{r_{a,t,k}^B}.$$

Away from baseline the same expression holds with the voluntary share evaluated at the scenario, $r^{V,S}/r^S$, rather than at baseline. The cap on $r^{V,S}$ also applies away from baseline; once the cap binds, the local total semi-elasticity is zero.

Two channels of heterogeneity emerge from one structure. The bracket $1 - M_{a,k}(c)$ reflects regime-driven variation in tax responsiveness: removing step-up shrinks the bracket and dampens elasticities. The voluntary share $r^{V,B}/r^B$ reflects asset-class variation: housing and pass-throughs have a high turnover share, so their total realization rate is mechanically less sensitive to rate changes than equities. Both effects show up in the same equation without additional parameters.

4 Death Treatment

Death is handled by three routing shares:

$$\delta_{\text{vanish}} + \delta_{\text{route}} + \delta_{\text{realize}} = 1, \quad \delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}} \in [0, 1].$$

Regime	δ_{vanish}	δ_{route}	δ_{realize}
Step-up basis	1	0	0
Carryover basis	0	1	0
Deemed realization	0	0	1

Only the routed share enters heirs' future unrealized gains. The death-flow contribution to next-period stock is

$$\Delta G_{h,t+1,k}^{\text{death}} = \delta_{\text{route}} \sum_a \omega_{a \rightarrow h,t} m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k}).$$

The deemed-realization share does not enter future stock. It becomes immediate death-event revenue:

$$R_{t,k}^{\text{death}} = \delta_{\text{realize}} \sum_a m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k}) \bar{\tau}_{a,t}.$$

$\bar{\tau}_{a,t}$ is the applicable average capital-gains rate for deemed realizations in the cell, after any regime-specific exemptions or rate rules.

5 Stock Law of Motion

The survivor channel is the main stock-flow equation. Survivors keep only the part of their gain stock that is not realized during the year. The reform-minus-baseline survivor contribution is

$$\Delta G_{h,t+1,k}^{\text{surv}} = \sum_a A_{a \rightarrow h,t} (1 - m_{a,t}) \left[(1 - r_{a,t,k}^S) \Delta G_{a,t,k} + G_{a,t,k}^B (r_{a,t,k}^B - r_{a,t,k}^S) \right].$$

The first term inside the brackets carries forward the policy-induced stock already present at the start of the year. The second term creates new stock difference because the reform changes this year's realization rate. If $r^S < r^B$, more gains stay locked in and ΔG rises. If $r^S > r^B$, the reform realizes more gains and drains the stock.

Adding survivor flow and death routing gives the full recurrence:

$$\begin{aligned} \Delta G_{h,t+1,k} = & \sum_a A_{a \rightarrow h,t} (1 - m_{a,t}) \left[(1 - r_{a,t,k}^S) \Delta G_{a,t,k} + G_{a,t,k}^B (r_{a,t,k}^B - r_{a,t,k}^S) \right] \\ & + \delta_{\text{route}} \sum_a \omega_{a \rightarrow h,t} m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k}). \end{aligned}$$

This is the core model. Given all age cells in year t , it computes all age cells in year $t + 1$ in one pass.

Three checks matter:

- If reform equals baseline, then $r^S = r^B$, $\delta_{\text{route}} = 0$, and ΔG remains zero.
- Under carryover, death moves unrealized gains to heirs rather than erasing them.
- Under deemed realization, death removes the stock and records revenue immediately.

6 Realizations and Revenue

The structural realization rate is applied as a ratio to observed baseline realizations:

$$R_{a,t,k}^S = R_{\text{obs}}^B(a, t, k) \frac{r_{a,t,k}^S G_{a,t,k}^S}{r_{a,t,k}^B G_{a,t,k}^B}.$$

Since $G^S = G^B + \Delta G$, the change in realized gains is

$$\Delta R_{a,t,k} = R_{\text{obs}}^B(a, t, k) \left[\frac{r_{a,t,k}^S (G_{a,t,k}^B + \Delta G_{a,t,k})}{r_{a,t,k}^B G_{a,t,k}^B} - 1 \right].$$

The ratio form is deliberate. It guarantees exact baseline reproduction when there is no reform: when $r^S = r^B$ and $\Delta G = 0$, the ratio is one and $R^S = R_{\text{obs}}^B$. In cells where $r_{a,t,k}^B G_{a,t,k}^B = 0$, use the standard sparse-cell fallback: either apply $r^S G^S$ directly or pool the baseline rate within asset class.

Revenue from ongoing realizations is

$$T_{a,t,k}^S = \tau_{a,t}^S R_{a,t,k}^S, \quad T_{a,t,k}^B = \tau_{a,t}^B R_{a,t,k}^B.$$

The reform-minus-baseline revenue change is

$$\Delta T_{a,t,k} = \tau_{a,t}^S R_{a,t,k}^S - \tau_{a,t}^B R_{a,t,k}^B.$$

Total revenue for year t adds ongoing realization revenue and any deemed-realization revenue at death:

$$\Delta T_t = \sum_{a,k} \Delta T_{a,t,k} + \sum_k R_{t,k}^{\text{death}}.$$

7 Calibration

The model has a small set of inputs. Most are observed or calibrated to baseline data. Two are judgmental: the carryover bequest parameter and the turnover share by asset class.

Object	Role	Source
$r_{a,t,k}^B$	Baseline realization rate	Simulator baseline
$m_{a,t}$	Mortality	Cell data or life table
η_k	Voluntary tax-price response	Calibrated by simulation
$\lambda_{a,t,k}^I$	Turnover hazard	Share of r^B by asset class
β	Discount factor	Default near 0.96
c	Death burden share in behavior	Regime rule
θ	Carryover bequest motive	Tunable assumption
$\omega_{a \rightarrow h,t}$	Heir age routing	Estate module

7.1 Turnover Hazard

The turnover hazard captures sales that are weakly tax-elastic: business exits, life-event home sales, divorce-induced asset divisions, forced liquidations, non-tax-driven portfolio rebalancing. The natural calibration is as a share of the baseline realization rate:

$$\lambda_{a,t,k}^I = \phi_k^I r_{a,t,k}^B, \quad \phi_k^I \in [0, 1).$$

The share form is automatically non-negative, automatically respects the ceiling $\lambda_{a,t,k}^I \leq r_{a,t,k}^B$, and degrades gracefully in sparse cells where r^B is small or zero. Allowing the turnover share ϕ_k^I to depend on age is a reasonable refinement but not pursued in the default calibration.

Three calibration paths are reasonable, in increasing order of effort:

1. **Default share by asset class.** Pick a single ϕ_k^I and use the share form above. Reasonable starting values: $\phi_k^I \approx 0.3$ for liquid equities, 0.7 for primary and other homes, 0.5 for pass-throughs and real estate funds.
2. **Implied from elasticity heterogeneity.** If the asset-class long-run target elasticity is ε_k and a common voluntary structural elasticity is $\varepsilon^V = -\eta(1 - M(0))$, the implied turnover share is

$$\phi_k^I = 1 - \frac{\varepsilon_k}{\varepsilon^V}.$$

This route makes cross-asset elasticity differences the explicit drivers of cross-asset turnover shares.

3. **External data.** Use mortgage origination and life-event-driven sale rates from real-estate panels for housing, business-exit rates from BEA / Census Business Dynamics Statistics for pass-throughs, and survey data on non-tax-driven portfolio rebalancing for equities. Convert the observed absolute hazard to a share by dividing by r^B , capped at one.

7.2 Tax-Price Curvature

The closed-form aggregate semi-elasticity expression

$$\left. \frac{\partial \log \sum_{a,t} R_{a,t,k}^S}{\partial \tau} \right|_{\tau=\tau_0, c=0} = -\eta_k (1 - M_{a,k}(0))_{\text{aggregated}} \frac{r_k^{V,B}}{r_k^B}$$

is the right intuition for what η_k does, but it is not a precise calibration target. With the cap on $r^{V,S}$, sparse-cell pooling, stock-feedback dynamics, and weighted aggregation, the actual aggregate elasticity of $\sum R^S$ with respect to τ differs from the closed form. Calibrate η_k by simulation: perturb τ by a small amount under current law, re-run the bathtub recurrence, and find the η_k that matches the target aggregate semi-elasticity. The voluntary-share scaling implies the simulation-calibrated η_k is larger in magnitude than in a single-channel calibration that hits the same target, because part of the baseline rate is held inelastic.

7.3 Death Burden Share

The default death burden share is

$$c = \begin{cases} 0 & \text{step-up,} \\ \theta & \text{carryover,} \\ 1 & \text{deemed realization.} \end{cases}$$

A reasonable starting value for carryover is $\theta = 0.5$, with sensitivity tests around it. The interpretation is simple: the holder behaves as if half of the heir's future capital-gains tax matters for the holder's own decision.

8 Cell-Level Conventions

For joint filers, assign the household to the older spouse's age:

$$a_{\text{joint}} = \max(a_1, a_2).$$

The joint-filer mortality rate is the probability that the joint-filer unit ceases to exist as a joint unit:

$$m_{\text{joint},t} = m_{1,t}m_{2,t}.$$

Within an age-year-asset cell, distribute $\Delta R_{a,t,k}$ to records in this order:

1. Pro rata to baseline realized gains, if the cell has baseline realized gains.
2. Pro rata to baseline unrealized gain stock, if baseline realizations are zero but gain stock is positive.
3. Assign nothing if both baseline realizations and baseline gain stock are zero.

The youngest age cell has no survivor inflow. It can receive unrealized gains only through inheritance under carryover. The topcoded age cell loops into itself.

9 Model Boundaries

The model is meant to capture realization timing, lock-in, carryover routing, and deemed-realization revenue. It does not try to solve every capital-gains problem.

Five limits should be kept in view:

- Accruals on existing assets are treated as policy-invariant, so they difference out of ΔG .
- The turnover / voluntary partition is a structural assumption. In reality the line is fuzzy: a divorce-driven home sale could be deferred a year if the tax cost is large enough. The model collapses that fuzziness into a single hazard λ^I that is treated as policy-invariant.
- Charitable transfers of appreciated assets are not modeled as a separate stock drain.
- Announced future tax rates can enter through the path τ_{t+j} , but unannounced future policy risk is not forecast endogenously.
- Mark-to-market regimes are outside this specification because annual taxation of accruals changes the revenue channel rather than just the realization timing channel.